

SHANI GUPTA

B.Tech Biotechnology | Bioinformatics | AI & Healthcare Research | Web Development | +91-9519340353 |
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PROFESSIONAL SUMMARY

B.Tech Biotechnology undergraduate with an **SGPA of 8.6**, passionate about biotechnology, bioinformatics, artificial intelligence in healthcare, and interdisciplinary research. I am skilled in **Biotechnology, Python development, and Web Development**, with hands-on experience in building web applications and solving technical problems. I am interested in integrating biology and technology to develop innovative healthcare solutions. Seeking internship and research opportunities where I can apply my technical, analytical, and problem-solving skills while contributing to impactful projects in biotechnology and AI.

EDUCATION

Rama University

Jun 2024 – May 2028

B.Tech Biotechnology (2024–2028) CGPA:8.6/10

Tilak Inter CollegeSchool

April 2023 – Feb 2024

Class XII (UP Board):79%

Sardar Patel Rashtriya InterCollege

April 2022 - Feb 2023

Class X (UP Board): 77%

PROJECTS

Protein Structure Explorer -

- Protein sequence analysis
- Molecular weight
- Amino-acid composition
- Hydrophobicity
- Secondary-structure analysis
- Python/Biopython
- Web-based visualization & Research

DNA Sequence Analysis -

- DNA → RNA transcription
- Translation
- GC content
- AT/GC ratio
- Reverse complement
- ORF detection
- Sequence statistics

AI-Based Drug Discovery Pipeline -

- SMILES → molecular descriptors
- Drug-likeness analysis
- Lipinski's Rule of Five
- Molecular fingerprints
- ML classification/regression

- Activity prediction
- Interactive visualization

PCR Primer Design & Optimization Platform -

- Target DNA sequence input & primer generation
 - Primer length, GC% & Tm analysis
 - Hairpin & dimer risk screening
 - Forward/reverse primer pair selection
 - Candidate comparison & ranking
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LEADERSHIP & EXPERIENCE

Google Developer Virtual Internship | EduSkills-AICTE | Oct 2025 - Dec 2025

- Completed Google technologies internship focused on development concepts to practical projects.

Python Developer Intern | EduSkills-AICTE | Jun 2026 - Aug 2026

- Completed Python development internship focused on programming, concepts to practical projects.

Campus Ambassador | Internshala & Neneev AI | Aug 2026 - Oct 2026

- Promoted internship and AI learning opportunities through student engagement, awareness campaigns.

Web Developer | BioGenius Club | Aug 2025 - Present

- Developed web applications & maintain and focused on frontend and Backend concepts to practical projects.
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TECHNICAL SKILLS

Programming: Python, R, JavaScript, C, Data Structures & Algorithms.

Bioinformatics: Biopython, Sequence Analysis, NCBI, Biological Data Analysis

Biotechnology: Molecular Biology, Microbiology, Genetics, Laboratory Techniques

AI/ML: NumPy, Pandas, Scikit-learn, TensorFlow

Tools: Jupyter Notebook, VS Code, Git, GitHub, Postman, Vercel

Database: MongoDB, Firebase

Soft Skills: Leadership, Communication, Teamwork, Problem Solving

CERTIFICATIONS & ACHIEVEMENTS

- Certificate of Excellence – Manipal Biotech Hackathon
 - NPTEL: Molecular Biology
 - NPTEL: Public Health
 - HP: AI & Machine Learning
 - Robotics & IoT Workshop
 - Python Developer Internship (EduSkills)
 - Google Developer Virtual Internship.
 - HP : Professional Networking for Career Growth
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